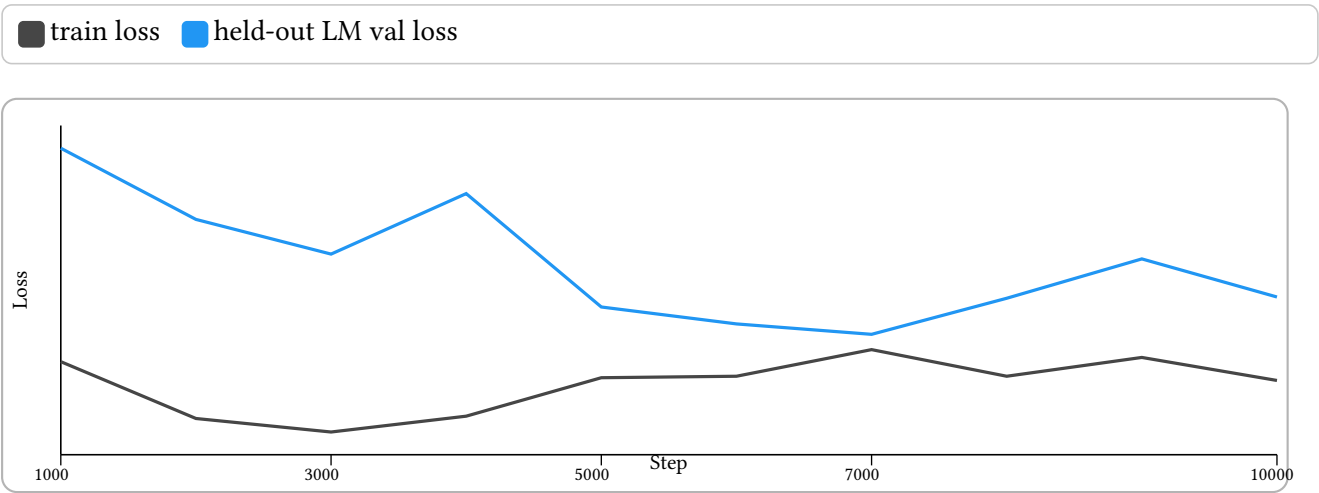


Long Grokking Probe, first 10k steps

Config: long_40m_ll_grokking_probe_100k.yml

Model: 38.64M params

Run: pure LM, 1x RTX 5090



Best held-out LM val loss in the first 10k is 1.3124 at step 7000.

- step 1000: val 2.1934
- step 3000: best train loss 0.8500
- step 7000: best val loss 1.3124
- step 10000: val 1.4891

This does not look like classic delayed grokking yet. The first 10k steps look more like ordinary early generalization improvement, with the best checkpoint appearing in the first ctx256 phase and then regressing slightly by 10k.

step	train	val
1000	1.1830	2.1934
3000	0.8500	1.6922
5000	1.1070	1.4418
7000	1.2400	1.3124
10000	1.0940	1.4891